



AUGMENTED ALZHEIMER MRI

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APRIL 29, 2026

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Abstract

Alzheimer’s disease is a progressive neurodegenerative disorder and one of the leading causes of dementia worldwide. Early and accurate detection is important for timely intervention, treatment planning, and improved patient outcomes. In this project, we develop and compare three deep learning approaches for multi-class Alzheimer’s disease classification using brain MRI scans. The study considers four categories: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. The evaluated models are a ResNet18-based architecture enhanced with attention, a region-aware ResNet combined with transformer attention, and an EfficientNet-B0 model with an additional attention module for computational efficiency. To ensure a fair comparison, all models were trained on the same subset of the Augmented Alzheimer MRI dataset using identical stratified train-validation-test splits and consistent preprocessing settings. Experimental results show that the ResNet18 + CBAM model achieved the best overall performance, while the region-aware transformer model and the EfficientNet-based model were less effective under the current data and training constraints. These findings suggest that lightweight attention integrated with a strong CNN backbone remains the most reliable choice for this dataset.

I. Introduction

Brain MRI is widely used in Alzheimer’s disease diagnosis because it can reveal subtle structural changes linked to disease progression, but interpreting these scans manually can be difficult and time-consuming. Deep learning has shown strong potential for MRI-based Alzheimer’s classification, but different model families offer different advantages. CNN-based architectures are effective at learning local spatial patterns, attention mechanisms help emphasize disease-relevant regions, and transformer-based approaches capture broader contextual relationships. At the same time, lightweight models remain important when computational resources are limited. Based on these differences, this study compares three deep learning strategies for Alzheimer’s disease classification from MRI images: an attention-enhanced CNN baseline, a region-aware hybrid model with transformer attention, and a lightweight EfficientNet-based model.

II. Literature Review

Recent deep learning studies on Alzheimer’s disease diagnosis suggest three main directions. First, CNN-based MRI classifiers remain strong baselines. Alsubaie et al. proposed an attention-driven CNN with multi-activation fusion and reported 92.1% accuracy and 0.99 AUC, showing that lightweight attention can still be highly effective for MRI classification. Second, hybrid image models that combine CNNs with Vision Transformers have shown strong results. Adeniran et al. presented an explainable MRI-based ensemble architecture integrating multiple CNNs with a Vision Transformer and reported 98.0% accuracy, suggesting that combining local feature extraction with global attention can improve both performance and interpretability. Third, multimodal fusion has emerged as a more advanced direction. Qiu et al. combined MRI with demographic and clinical information across eight datasets, while Rahman et al. extended this idea through NeuroNet-AD, which fused MRI with structured metadata using ResNet-18, CBAM, BERT, and cross-attention, reporting 98.68% accuracy, 99.40% recall for AD, and 0.98 AUC. Together, these studies suggest that multimodal fusion can outperform single-modality systems when the fusion strategy is carefully designed. Related work in cancer imaging and multimodal mental-health analysis also supports the use of attention, region-aware preprocessing, and hybrid architectures, which helped motivate the current study.

III. Methodology

A. Dataset and Preprocessing

The experiments were conducted on the Augmented Alzheimer MRI dataset, sourced from Kaggle, which contains four classes: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented which represent the progressive stages of Alzheimer’s disease, making the task a multi-class classification problem. To ensure fair evaluation and preserve class distribution, stratified sampling was used for the train-validation-test split, and a subset of 4,000 images was selected due to CPU-based training constraints; the dataset split and preprocessing pipeline are summarized in Tables I and II.

Table I:

Dataset	Percentage	No. of Samples
Training	70%	2800
Validation	15%	600
Testing	15%	600

Table II:

Step	Description
Resizing	Images resized to 128x128
Normalization	Pixel values scaled to $[-1,1]$
Data Augmentation	Random horizontal flip and rotation ($\pm 10^\circ$)
ROI Extraction (Model 2 only)	Threshold-based masking to focus on brain region.

B. Proposed Models and Hyperparameters

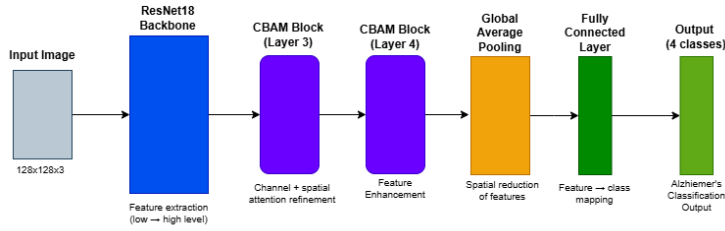
Three deep learning architectures were implemented to compare different design strategies for Alzheimer’s MRI classification. The first model, ResNet18 + CBAM, serves as the baseline and uses lightweight attention to refine CNN feature extraction, followed by global average pooling and a fully connected classifier.. The second model, a Region-Aware ResNet + Transformer, extends this idea by incorporating ROI-based preprocessing and transformer attention to capture broader contextual relationships. The third model, EfficientNet-B0 + Attention, was designed as a lighter alternative that aims to balance classification performance with computational efficiency. The main architectural components and training settings of all three models are summarized in Table III.

Model	Backbone	Attention / Extra Module	Key Design	Classifier	Learning Rate	Optimizer	Loss Function	Batch Size	Epochs
ResNet18 + CBAM	ResNet18 (pretrained)	CBAM after Layers 3 and 4	Baseline CNN with attention-enhanced feature refinement	Global average pooling + FC + Dropout (0.4)	3e-5	AdamW	Weighted Cross-Entropy	8	12
Region-Aware ResNet + Transformer	ResNet18	Channel attention + Transformer (2 layers, 2 heads)	ROI extraction + 1x1 conv (512→128) + transformer-based global modeling	FC + Dropout (0.5)	3e-5	AdamW	Weighted Cross-Entropy	8	12
EfficientNet-B0 + Attention	EfficientNet-B0 (pretrained)	CBAM-style attention block	Lightweight architecture for computational efficiency	2 FC layers + Dropout	2e-5	AdamW	Weighted Cross-Entropy + Label Smoothing (0.1)	8	12

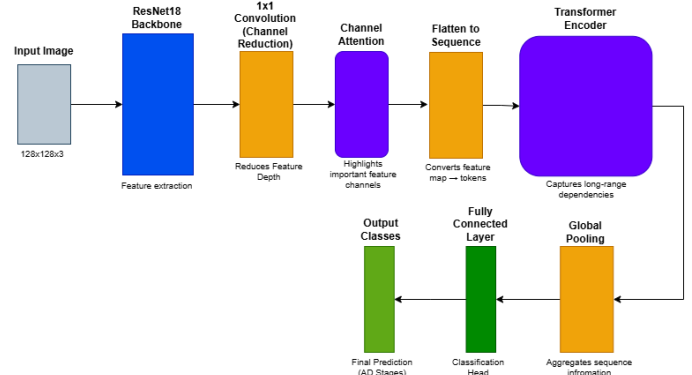
C. Workflow and Architecture Diagrams

Model Architecture Diagrams:

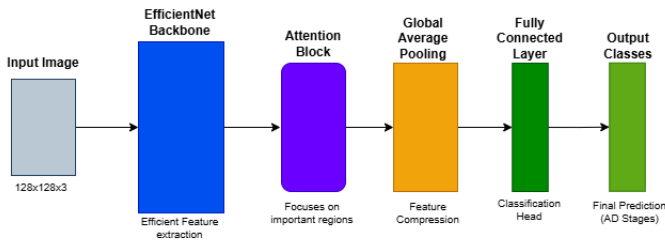
Approach 1: ResNet + CBAM:



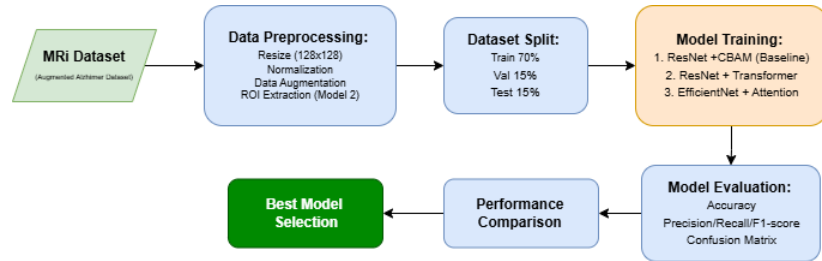
Approach 2: CNN + Transformer:



Approach 3: EfficientNet + Attention:



Overall Workflow Diagram:

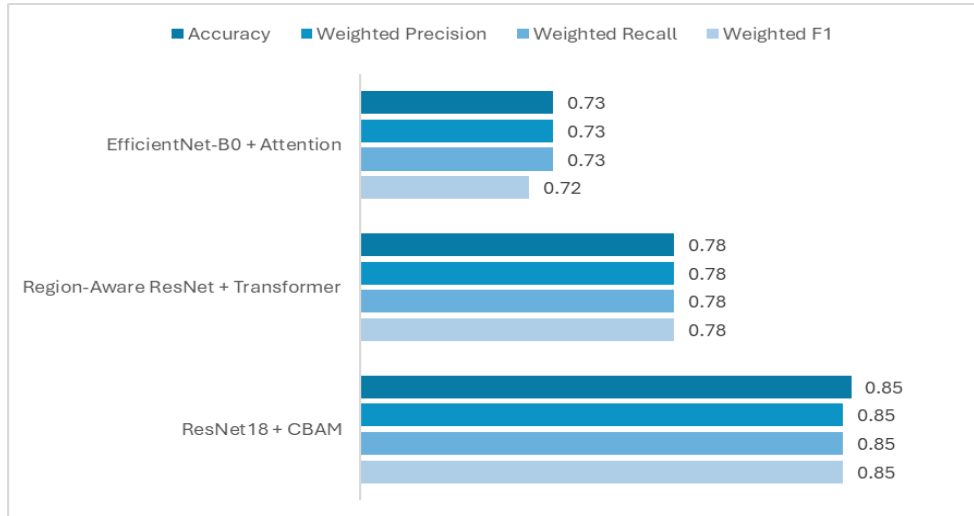


IV. Evaluation and Result Analysis

All three models were evaluated on the same held-out test set of 600 images using accuracy, weighted precision, weighted recall, and weighted F1-score, with confusion matrices used for class-level analysis.

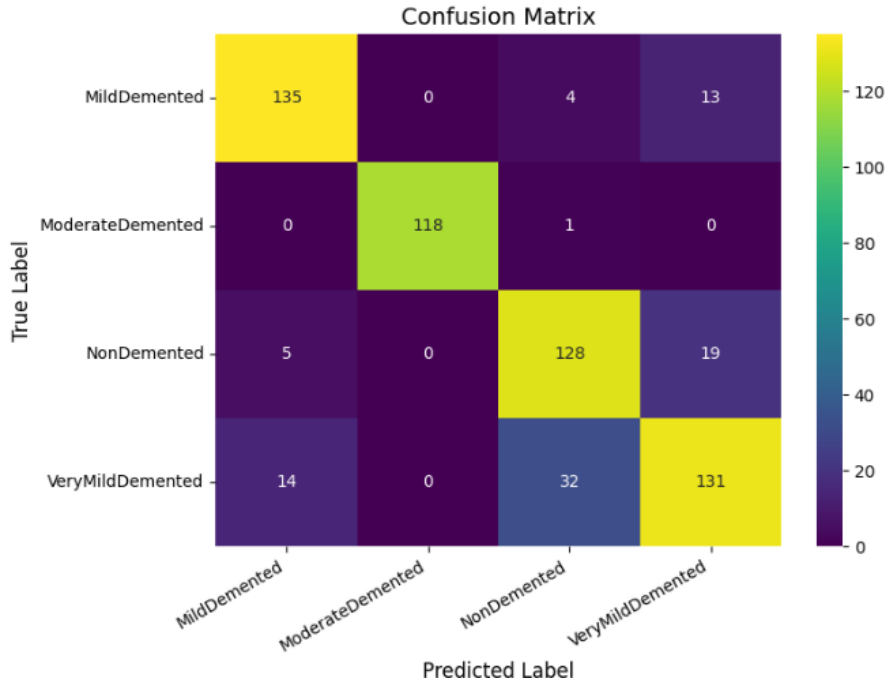
A. Quantitative Results

Performance Comparison of the Proposed Models:



The ResNet18 + CBAM model achieved the best overall performance across all reported metrics, followed by the Region-Aware ResNet + Transformer model and EfficientNet-B0 + Attention.

Confusion matrix of the ResNet18 + CBAM model on the test set:



B. Discussion

ResNet18 + CBAM achieved the best overall performance, showing that a standard CNN base model with lightweight attention was better suited to the current MRI dataset than the more complex alternatives. The Region-Aware ResNet + Transformer model did not outperform the baseline, likely because transformer-based attention is more sensitive to limited data and to the loss of spatial structure after feature sequence conversion. The EfficientNet-based model produced the weakest results, which may reflect domain mismatch from ImageNet pretraining and stronger regularization under short CPU-based training.

C. Comparison with Literature

Study / Model	Main Approach	Reported Performance	Comparison to This Study
Alsubaie et al.	Attention-driven 3D-CNN with multi-activation fusion	92.1% accuracy, 0.99 AUC	Higher than our best model; likely due to stronger setup and optimized MRI pipeline
Adeniran et al.	Explainable CNN + ViT ensemble	98.0% accuracy	Higher than our transformer-inspired model; their ensemble is more advanced and computationally heavier
Qiu et al.	Multimodal MRI + clinical fusion	Strong multi-metric performance across 8 datasets	Stronger generalization than our image-only models due to multimodal fusion
Rahman et al.	NeuroNet-AD multimodal MRI + metadata	98.68% accuracy, 0.98 AUC	Outperforms all our models; supports future extension toward multimodal Alzheimer diagnosis
<i>This study:</i> ResNet18 + CBAM	CNN + lightweight attention	85.33% accuracy, 0.85 weighted F1	Best model among the three tested approaches
<i>This study:</i> Region-Aware ResNet + Transformer	ROI-based CNN + transformer attention	78.00% accuracy, 0.78 weighted F1	Moderate performance, but lower than the baseline
<i>This study:</i> EfficientNet-B0 + Attention	Lightweight transfer learning + attention	73.00% accuracy, 0.72 weighted F1	Lowest-performing model in this study

The results are broadly consistent with the literature. CNN-based attention models remained strong for MRI classification, while transformer-based and multimodal approaches showed promise but generally depended on stronger data conditions, richer fusion strategies, or more advanced optimization than those used in this study. These findings suggest that future improvements should focus on multimodal fusion with clinical metadata.

V. Conclusion and Future Work

In this study across the tested models, ResNet18 + CBAM achieved the best overall performance, showing that a strong CNN architecture with lightweight attention was more effective on the current dataset than the transformer-based and EfficientNet-based alternatives. The region-aware transformer model did not outperform the baseline, likely because the available data and preprocessing pipeline were not sufficient to fully support transformer-based global modeling. The EfficientNet-B0 + Attention model provided a lighter solution, but its performance remained below the other two models. Overall, the results suggest that matching the model design to the data characteristics is more important than increasing architectural complexity alone. Future work should focus on stronger augmentation, longer training, external validation on datasets such as ADNI or OASIS, multimodal fusion with clinical metadata, and completion of the full agentic pipeline for automated report generation and user interaction.

VI. Agentic AI Component

This project is designed to extend beyond classification by linking the predicted Alzheimer stage to an automated explanation and reporting workflow. A prototype demonstrating this agentic component is available through the Claude link.

Demo link: <https://claude.ai/public/artifacts/859d5d5e-6116-4eca-8caa-012bd2e7d75b>

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